Candidate Name	Centre Number	Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

CHEMISTRY **4024/3**

PAPER 3 Practical Test — 1 hour 30 minutes

SPECIMEN PAPER

Candidates answer on the question paper.
Additional materials:
As listed in Instructions to Supervisors
Scientific calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of each question or part question.
Qualitative analysis notes for this paper are printed on page 7.

FOR EXAMINER'S USE	
1	
2	
TOTAL	

This specimen paper consists of 7 printed pages.

1. **(a)** You are required to determine the concentration of a solution, **S**.

Solution **S** is aqueous potassium hydroxide, KOH.

Solution **R** is 1.00 $moldm^{-3}$ sulphuric acid, H_2SO_4

Fill the burette with **R**.

Transfer 10.00 cm^3 of **R** into a polystyrene cup.

Measure and record, the initial temperature of **R** in **Table 1.1**

Measure 40.00 cm^3 of **S** using a measuring cylinder.

Add this to the polystyrene cup containing **R**.

Stir the solution using a thermometer.

Measure and record the highest temperature reached in **Table 1.1**

Repeat the procedure using volumes of **R** and **S** shown in **Table 1.1**

Table 1.1

volume of R /cm^3	**volume of S/ cm^3**	**initial temperature of R/ ^oC**	**highest temperature of mixture/ ^oC**	**temperature rise/ ^oC**
10	40			
20	30			
30	20			
40	10			

Complete **Table 1.1**. [12]

(b) Plot a graph of temperature (*y-axis*) against volume of **R** (*x-axis*).

[3]

(c) **(i)** Deduce the highest temperature rise.

....

[1]

(ii) Determine the volumes of **R** and **S** that produced the highest temperature rise.

*1. volume of **R*** ....

*2.volume of **S*** ....

[2]

(d) **(i)** Write a balanced chemical equation for the reaction between **R** and **S**.

....

.... [1]

(ii) Calculate the number of moles of **R** at the highest temperature.

[1]

(iii) Determine the concentration of **S** in $moldm^{-3}$.

[1]

2. You are required to identify the chemical composition of a chemical, **C.**

Carry out the test(s) described in **Table 2.1**.

Table 2.1

(a)

test(s)	**observation(s)**	**deduction(s)**
(a) describe the appearance of chemical **C**		
(b) dissolve a spatula of **C** in about 20.0 cm^3 of distilled water in a test tube *use the solution for tests **(c)(i)** and **(c)(ii)***		
(c) (i) to a portion of **C**, add aqueous sodium hydroxide until in excess		
(ii) to another portion of **C**, add aqueous ammonia until in excess filter the mixture and divide the filtrate into **three** equal portions. Use these portions for test **(d) (i)**, **(ii) and (iii)**		

test(s)	observation(s)	deduction(s)
(d)(i) to the first portion add dilute nitric acid followed by silver nitrate,		
(ii) to the second portion add dilute nitric acid. followed by barium nitrate		
(iii) to the third portion add aqueous sodium hydroxide, warm the mixture		

[18]

Complete the **Table 2.1** by stating the observations and the deductions made for each test.

(b) The ions present in chemical **C** are

1. cations ....

2. anions ....

[2]

QUALITATIVE ANALYSIS NOTES

Tests for anions

anion	*test*	*test result*
carbonate (CO_3^{2-})	add dilute acid	effervescence, carbon dioxide produced
chloride (Cl^-) [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	white ppt.
iodide (I^-) [in solution]	acidify with dilute nitric acid, then add aqueous lead (II) nitrate	yellow ppt.
nitrate (NO_3^-) [in solution]	add aqueous sodium hydroxide then aluminium foil; warm carefully	ammonia produced
sulphate (SO_4^{2-}) [in solution]	acidify with dilute nitric acid then add aqueous barium nitrate.	white ppt.

TESTS FOR AQUEOUS CATIONS

cation	*effect of aqueous sodium hydroxide*	*effect of aqueous ammonia*
aluminium (Al^{3+})	white ppt., soluble in excess colourless solution	white ppt., insoluble in excess giving a
ammonium (NH_4^+)	ammonia produced on warming	
calcium (Ca^{2+})	white ppt., insoluble in excess	no ppt.
copper (Cu^{2+})	light blue ppt., insoluble in **excess**	light blue ppt., soluble in **excess giving a dark blue solution**
iron (II) (Fe^{2+})	green ppt., insoluble in excess	green ppt., insoluble in excess
iron (III) (Fe^{3+})	red-brown ppt., insoluble in	red-brown ppt., insoluble in excess excess
zinc (Zn^{2+})	white ppt., soluble in excess giving a colourless solution	white ppt., soluble in excess giving a colourless solution

TESTS FOR GASES

gas	***test and result***
ammonia (NH_3)	turns damp red litmus paper blue
carbon dioxide (CO_2)	turns limewater milky
chlorine (Cl_2)	bleaches damp litmus paper
hydrogen (H_2)	“pops” with a lighted splint
oxygen (O_2)	relights a glowing splint
sulphur dioxide (SO_2)	turns aqueous potassium dichromate (VI) from orange to green